

eGlove: Designing Interactive Fabric Sensor for Enhancing Contact-Based Interactions

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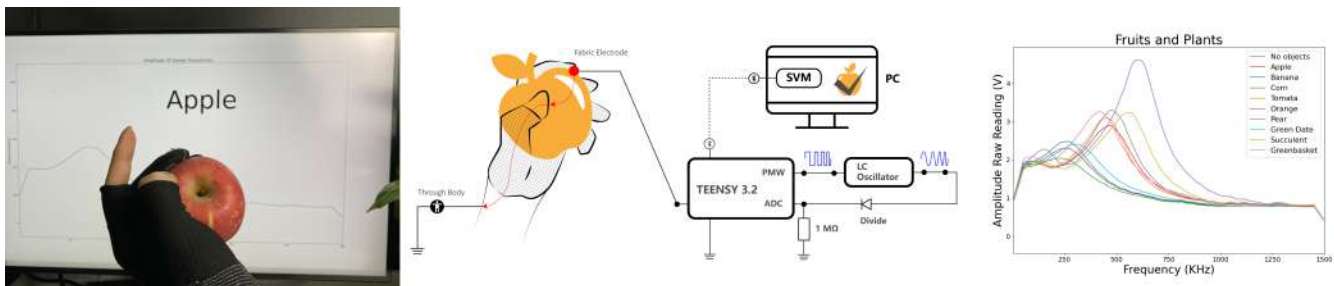


Figure 1: (a) eGlove is an interactive fabric sensor for recognition of different human body parts and objects such as an apple; (b) The sensing principle of our system; (c) The amplitude responses of different fruits and plants using Swept Frequency Capacitive Sensing.

ABSTRACT

We present *eGlove*, a wearable and low-cost fabric sensor for recognizing a rich context of objects by touching them, including daily necessities, fruits, plants, as well as different body parts. Our sensing approach utilizes Swept frequency Capacitive Sensing (SFCS) to provide consistent sensor readings even when the fabric electrode is under varying deformation and stretching degrees. Our work proposes an easy fabrication method and hardware configuration for prototyping the interactive fabric sensor. We evaluated our system's classification accuracy through per-user training and found a real-time classification of 96.3%. We also demonstrated novel contextual interactions enabled by our technical approach with several applications.

CCS CONCEPTS

• Human-centered computing → Interaction techniques.

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CHI '21 Extended Abstracts, May 8–13, 2021, Yokohama, Japan

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ACM ISBN 978-1-4503-8095-9/21/05...\$15.00

<https://doi.org/10.1145/3411763.3451824>

KEYWORDS

Interactive Fabrics; Swept Frequency Capacitive Sensing; Object Recognition; Contact-Based Interactions

ACM Reference Format:

Hebo Gong, Zhitong Cui, Yanan Wang, Chenyi Shen, Deyin Zhang, and Shijian Luo. 2021. *eGlove: Designing Interactive Fabric Sensor for Enhancing Contact-Based Interactions*. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '21 Extended Abstracts)*, May 8–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3411763.3451824>

1 INTRODUCTION

Fabrics have been one of the most commonly used materials in people's everyday life. In the area of Human-Computer Interaction, a collection of researchers and designers have explored interactive fabrics in a variety of applications such as clothing, furniture, and daily necessities [11, 20]. More recently, researchers and designers have explored interactive fabrics by enabling the contextual awareness for object recognition [4, 14, 24] and posture recognition [19, 25].

The ability of "smart" fabrics to recognize what is happening to them in the physical world would truly augment human activities [17]. However, it is difficult for sensing techniques to produce consistent electrodes' readings, especially when the fabric is deformed, potentially introducing false detection after the machine learning classifier has been trained. Therefore, we see an opportunity to enhance the ability of "smart" fabric not only to recognize what is

contacting with them but also to preserve the essential properties of the fabric. In this work, we sought to design a soft and elastic fabric sensor that could be worn on the body for contextual interactions while still maintaining the fabric's flexibility, deformability, and wearability. We utilized Swept Frequency Capacitive Sensing (SFCS) [12] for contact-based and robust recognition of different objects and human body parts. Our approach can provide consistent sensor readings even when the fabric is under varying deformation and stretching degrees.

We presented a proof-of-concept prototype called *eGlove*, which is worn on the hand. *eGlove* uses conductive fabric (area=1cm²) as the only electrode to emit an alternating current signal over a frequency range. We first tested the properties of the conductive fabric through four aspects *Size*, *Shape*, *Deformability* and *Stretchability*, to maximize the sensitivity of the signal. Then we manually fabricated the multi-layer fabric sensor by integrating conductive fabric with adhesive fabric and normal fabric. At last, we customized a low-cost sensing board and trained a machine learning model for classification. We tested our prototype using 10 daily necessities, 9 fruits and plants, and 6 body parts. Then we found a real-time classification accuracy of 96.3%. Our results demonstrated that *eGlove* is able to recognize objects with a certain impedance range, such as *daily necessities, fruits, plants, animals, humans* as well as different *body parts* for potential posture recognition in the future.

2 RELATED WORK

Our work is motivated by prior research in two domains, including input on interactive fabrics and capacitance and impedance Sensing.

2.1 Input on Interactive Fabric

With the current state-of-art, research on the input on interactive fabrics has been focused on touch [1, 9, 11, 23], deformation gestures [10, 18, 20], object recognition [14, 24] and posture recognition [25]. The enabling techniques behind these innovations can be divided into capacitive sensing [23, 24], impedance sensing [7, 10], optical sensing [6, 18] and inductive sensing [4].

For touch input on interactive fabrics, Musical Jacket [9] is an early exploration in this area. A user could interact with the jacket through an embroidered-capacitive touch keypad. Project Jacquard [11] describes a standard fabrication of conductive yarn which can distinguish whether the user touches or not. ZebraSense [23] is a dual-sided woven touch sensor that can differentiate interactions on the top and bottom surfaces of it.

For detecting deformation gestures, SmartSleeve [10] used a resistive-pressure textile to detect deformation gestures such as twirl, twist, fold, and push gestures. Glauser et al.[3] fabricated soft, capacitive sensor arrays that capture surface deformation. StretchEBand [20] allowed stretch-based interaction with wearables, mobile devices, and clothing using stitch-based stretching sensors.

For recognizing objects, Rofouei et al. [14] used a fabric sensor to recognize different objects by measuring the pressure footprint. Tessutivo [4] recognized metallic objects on spiral-shaped coils made of conductive thread using inductive sensing. Capacitivo [24] presented a capacitive object recognition technique on fabrics that

could detect the shape of the object's footprint and the object's material.

Compared with previous research, we focused on contextual interactions on wearable interactive fabrics. Our work complements existing research by recognizing different objects and body parts even when the fabric is under varying degrees of deformation and stretching.

2.2 Capacitive and Impedance Sensing

Capacitive and impedance sensing is widely employed in prior research for numerous applications. Most research in this field utilized capacitive sensing to recognize touch input [1, 9, 11, 23], 3D gestures [13, 26] and postures [19, 25]. DiamondTouch [1] created a contact-based technique to detect multiple users' input in the meantime. Rekimoto [13] introduced two devices which enable users to interact with wearable or nearby device by capacitive sensing. Electrick [27] enables touch input on irregular objects and surfaces using the electric field.

Human tissue bio-impedance sensing has been used for human health products such as digital scales and fluid composition measurement [2, 8]. Impedance sensing techniques have also been explored in HCI. Tomo [26] applied Electrical Impedance Tomography (EIT) to recognize wrist and hand gestures by measuring the interior impedance of eight electrodes. Zensei [15] measured the impedance through the human body tissue between pairs of electrodes. ThreadSense [7] locates touches on an interactive thread using impedance sensing. Compared with the above work that uses a single-frequency AC signal, Swept Frequency Capacitive Sensing could acquire much more information about an object's internal composition. Touché [12] captures the body's capacitance using a single sensing electrode to make objects touch and grasp sensitive and recognize bimanual hand gestures. Based on this approach, Harrison et al. [5] proposed capacitive fingerprinting to differentiate multiple touch-screen users. Our work explored how a similar sensing principle can be applied to wearable interactive fabrics to allow contact-based interactions.

3 SENSING PRINCIPLE

Alternating current (AC) can pass through dry human skin (~100KΩ) [16] or high resistance objects such as fruits, but direct current (DC) cannot. When the AC signal passes through a body-contacted object, a capacitive link is formed between electrode and ground through the object, skin, tissues, and bones [2, 21, 28]. Based on the fundamental observation that every body-contacted object and human body part have different tissues or biological factors, the amplitude and phase of the applied AC signal would change to varying degrees when it passes. The opposition of impedance properties causes this.

The electrical properties of AC signals are affected by two of the most prominent factors. Specifically, capacitive properties are affected by how objects contact the electrode, such as the contact area of the electrode and the human body. Besides, impedance properties are strongly affected by the signal frequency. Because AC signal will always flow through the path of least impedance inside the human body or organic objects, e.g., fruits and plants [2]. When the frequency of the applied AC signal changes, the

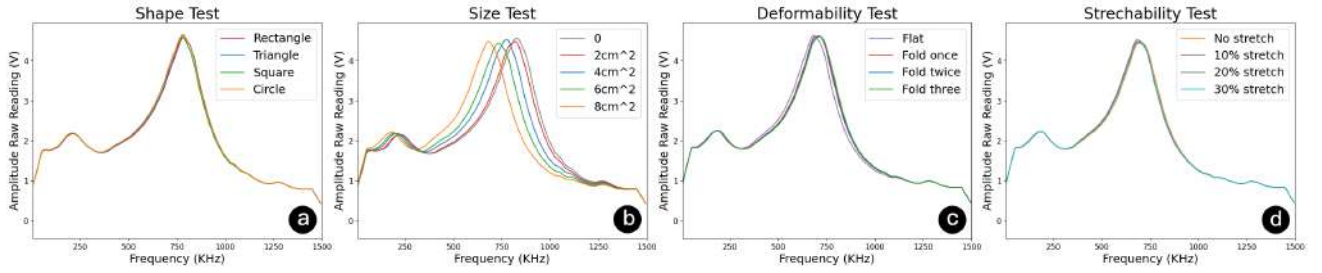


Figure 2: (a) The amplitude response of different shapes of fabric electrodes; (b) The amplitude response of different sizes of fabric electrodes; (c) The amplitude response of different deformability of fabric electrodes; (d) The amplitude response of different stretchability of fabric electrodes

electrical current flows different paths. This is because the electrical impedance of tissues or substances is frequency-dependent [5].

Therefore, we can distinguish between different objects and human body parts by comparing amplitude changes in AC signals over a wide range of frequencies. These changes can be measured in real-time and used to build a frequency-to-impedance profile. Furthermore, the advantage of SFCS is that only a single electrode and wire are needed, and we can obtain a wealth of information about the internal tissue composition of an object or a human body part. We exploit this information for classification. It is worth noting that the target objects for recognition need to be exposed to the electrode and the human body simultaneously, so that current can pass through the object and the human body connected to the common ground to form a loop (Figure 1-b), as the sensing principle only applies to the grounded situation.

4 IMPLEMENTATION

In this section, we articulate the design of *eGlove* in detail, hardware, including the design of the interactive fabric sensor, software, and the technical evaluation of our system.

4.1 Hardware

Our customized sensing board comprises a signal generation module and a signal transformation module. The signal generation module consists of Teensy 3.2, Bluetooth, and a battery (3.3V). We found Teensy 3.2 can only generate square waves above 100 KHz. Thus, we used Teensy 3.2 to generate an evenly stepped square wave sweeps from 100 KHz to 1.6 MHz in a 5 KHz step (300 steps in each sweep). The return voltage signal is measured by a 1 MΩ resistor, and the sampling rate is up to 200 FPS. We also found that Arduino Uno (ATmega328p) can not produce an evenly stepped sweep signal so that the frequencies of PWM signals are too few to support us in detecting the desired high-frequency characteristics for our recognition, especially in the frequency range above 500 KHz.

The signal transformation module consists of an LC oscillator to transform the square wave signal into a sine wave signal. The frequency is controlled by Teensy 3.2, but the conversion efficiency of various frequency signals is affected by the electronic components' parameters. We can obtain the best frequency (f) square-to-sine conversion efficiency by changing capacitance (C) and inductance

(L) through the following formula. We chose electronic components for the most critical frequency range from 300 KHz to 1 MHz for our desired contextual interactions.

$$LC = \frac{1}{4\pi^2 f^2}$$

The total power of the entire hardware system is about 200 mW. With a 600 mAh battery, the system can keep working for nearly 6 hours. All units are available online for a total price of no more than 25 dollars.

4.2 Preliminary Test

We tested the properties of the conductive fabric through four aspects: size, shape, deformability, and stretchability. We expect to maximize the sensitivity of the signal through fabric sensor designing.

Shape. The electrodes (4cm^2) of the rectangle, triangle, square and circle were not significantly different in the case of swept frequency capacitive sensing (Figure 2-a), indicating that the shape of the electrode does not affect the measurement process.

Size. There is a clear trend that as area of electrode increases (square), from 4cm^2 , 16cm^2 , 36cm^2 to 64cm^2 , the entire curve shifts to the left as the size increase (Figure 2-b). Considering that touching objects will cause the highest peak of the curve to move to the left, this phenomenon means that the smaller electrode, which has a lower impedance, allows the impedance to be measured in a wider range. Therefore we should make the electrode area as small as possible while ensuring the target is in full contact.

Deformability. We contracted the response curves of the fabric electrodes that were flat, folded once, folded twice, and folded three times (Figure 2-c). No noticeable difference was found, which means the measurement results would not change even if the electrode has been deformed.

Stretchability. A rectangle fabric electrode was stretched from 0, 10%, 20% to 30%, and the measurement difference can be ignored (Figure 2-d). This means the electrode of this system can withstand a certain degree of stretching.



Figure 3: (a) The multi-layer structure of *eGlove*; (b) Adhering different layers by using an electric iron; (c) The outer side of the *eGlove*; (d) Connecting the conductive fabric with the wires through sewed conductive thread.

4.3 Fabrication Procedure

Based on the four properties test of the conductive fabric, we designed *eGlove* into a seven-layer structure (see Figure 3-a), including two parts: 1) an outer side of the glove for gathering the signal, and 2) the inner side acting as the shielding layer for effectively shielding the fabric sensor from the noise of human body. All the materials we used are off-the-shelf and can be purchased from the online shop.

The top layer is the elastic conductive fabric ($0.62\Omega/\text{cm}$) that was cut into 1cm^2 for emitting the signal; the fifth layer is the same conductive fabric that was cut into 1.5cm^2 for shielding the noise; the hot melt adhesive fabric (second, fourth and sixth layer) are sandwiched for combining each layer tightly by using a handheld electric iron (Supor, YD18A07J/T-15), see Figure 3-b. First, we fabricated the outer side of the glove by adhering to the smaller conductive fabric at the middle finger of the glove (Figure 3-c). Then we adhered the larger conductive fabric and regular fabric in order to create the shielding layer at the inner side of the glove. Finally, we manually sewed the conductive thread ($6\Omega/\text{cm}$) to the smaller conductive fabric for connecting with the wires and sensing board (Figure 3-d). The whole fabrication processes were low-cost and easily manipulated within a lab environment.

4.4 Software

We trained a machine learning model for classification using a pre-collected database of labeled references. This enabled rapid testing, iteration, and development of our contact-based interactions. And the fabric sensor reports 300 points of frequency-to-amplitude values to a conventional computer. We firstly used a moving average filter (sliding window, size=10) to reduce the fluctuations in sensor readings. Then we calculated the average of 30 samples and sent it to the classifier, which increased robustness to global variations caused by the harmonic wave. The final real-time sampling rate exceeds 64 FPS.

After data processing and initial tests, we derived a series of features for classification. We used all the 300 amplitude values from the frequency sweep as features. Then we extracted statistical features from the signal: mean, min, max, standard deviation, and the n frequency points with the highest peaks ($n=5$). To describe the overall shape of the signal profile, we computed the 1st derivative by down-sampling the data points into an array of 60 points as features. These features form the input to a SMO-based SVM

($C=10$, polynomial kernel, $\gamma=\text{scale}$, normalized) for real-time classification. Additionally, the hyper-parameters of the kernel are tuned through a grid search to improve accuracy. Then we applied 10-fold cross-validation to avoid model overfitting.

4.5 Technical Evaluation

We conducted a preliminary experiment to evaluate the classification accuracy of objects recognition and body parts recognition. Since the impedance properties of a particular human body can be fairly unique [5], our validation procedure employed a per-user classifier paradigm. We recruited 3 participants (1 female) for a live user study in our lab and trained different machine learning models for each recognition context. Each model was trained per-participant. After the participant put on the *eGlove*, we collected 200 data instances per objects and trained a model in-situ. Once trained, the participant was asked to randomly grasp different objects or touch different body parts. Real-time per-user recognition accuracy was recorded for analyzing the results.

Daily Necessities Recognition. Our technique can make daily necessities touch or grasp sensitive without instrumenting them with an accuracy of 97.4% (SD=3.5). We tested our prototype using 10 objects commonly found in households (Figure 4-b, Figure 4-c). The tested objects vary in exterior material properties and built-in substances or metallic components. Meanwhile, we found that our system could also distinguish whether there is water in a stainless steel cup. However, there is no obvious amplitude response for an empty plastic cup, which means we can not identify it.

Fruits and Plants Recognition. Aside from recognizing inanimate objects, our approach can also classify fruits and plants with an accuracy of 98.2% (SD=2.2). We tested our prototype using 7 fruits and 2 plants (Figure 4-a, Figure 1-c). This result reveals that we can also identify different animals.

Body Parts Recognition. Our sensing approach can also be used to identify the different body parts that are in contact with the conductive fabric electrode with an accuracy of 93.2% (SD=5.1). When a standing user touches different body parts, the impedance between the electrode varies as the signal flows through slightly different paths inside the user's body. Since the AC signal could pass through clothes, our approach could work even when the participant is wearing touch clothes. We tested our prototype with ear, chin, chest, elbow, waist, and knee, and with touching clothes on the

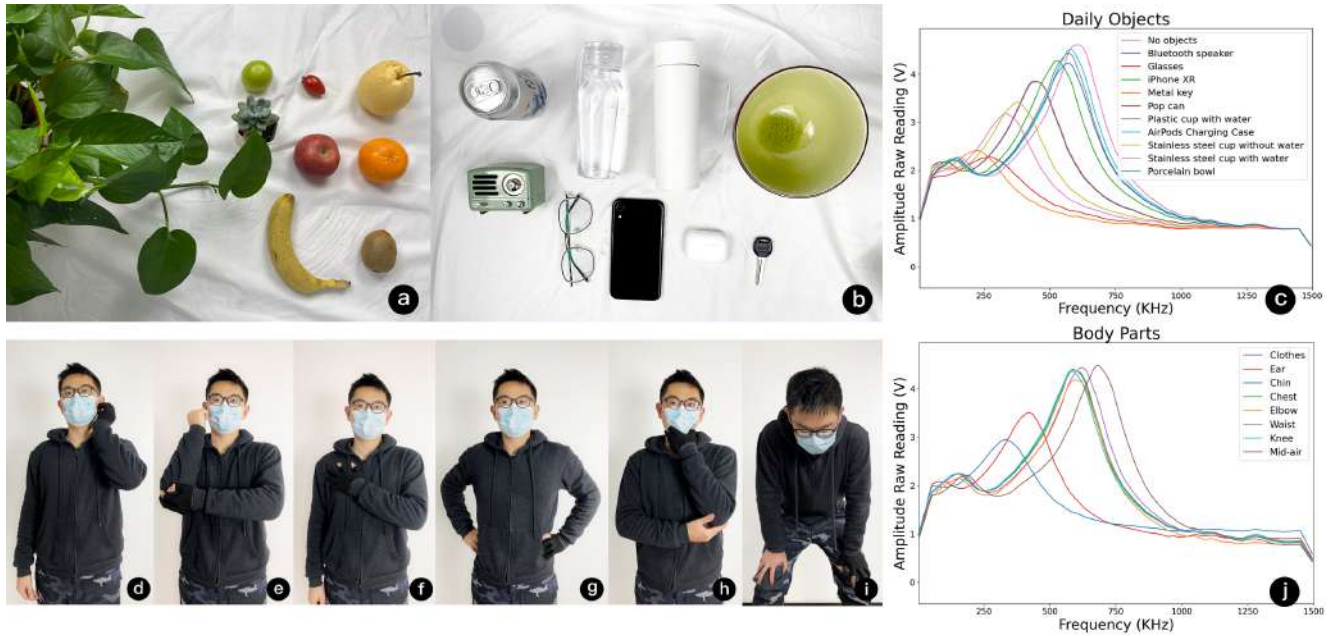


Figure 4: (a) The full list of our tested fruit and plants, including green basket, green dates, tomatoes, pears, succulents, apples, oranges, bananas, kiwis; (b) The full list of tested daily necessities, including pop can, full-water plastic cup, stainless steel cup with water, stainless steel cup without water, bowl, speaker, glasses, iPhone XR, AirPods charging case, key; (c) The amplitude responses of different daily necessities; (d~i) The full list of tested body parts, including ear, elbow, chest, waist, chin and knee. (j) The amplitude responses of different body parts and control groups.

desk and mid-air were set up as control groups (Figure 4). Although accuracy would be much lower without targeted per-user training because of the different clothes, it is worth noting that this result implicates potential posture recognition in the future.

5 ENVISIONED APPLICATIONS

Our technique could be used for recognizing grasped objects and touched body parts, which could potentially launch context-relevant functionalities automatically in various scenarios. We envisioned a series of applications that *eGlove* could be weaved into everyday life. For instance, the light will be turned on automatically when the user takes out the keys to open the door; after entering the house, he could easily start the coffee machine and get a cup of warm coffee by grabbing an empty ceramic coffee cup; he could also be able to simply start the Bluetooth speaker to play the music on his phone by touching his phone and the speaker in order.

People set many agendas for a better life, however it's difficult to stick to them. For example, people often forget to drink enough water or take medicine on time. The system could detect and recognize the user's meal or medicine every day, and report non-compliance through mobile phones or smart speakers (Figure 5-a).

In the case of severe COVID-19, it's necessary for people to minimize the number of times of touching the mouth and the nose. *eGlove* could effectively record the touched location and the number of touching times, and would kindly remind the users to pay attention to hygiene, such as changing the facial mask (Figure 5-c).

We also envisioned that *eGlove* could be an educational tool for assisting children's learning. A smart speaker could be involved in

this case. For example, as shown in Figure 5-b, the children could learn from touching the succulent, the system will represent the related information of the succulent automatically, such as its scientific knowledge or a relevant story.

6 LIMITATIONS AND FUTURE WORK

In this section, we discuss the insights derived from this work, the limitations of our technical approach, and future research directions.

Form Factors. As we discussed before, the electrode area's size should be as small as possible because the conductive fabric's impedance will increase as the area increases. We are planning to use conductive threads instead of conductive fabrics. By weaving the wire in different ways, we can reduce the electrode's impedance while ensuring sufficient contact area. Besides, we only measured the amplitude of the SFCS signals through one electrode in this approach. It is possible to get richer information if we measure the phase of the signal or apply multi-electrode equipment in the future. In addition, the LC oscillation circuit reduces the hardware cost, but the signal generation module should be redesigned if a more accurate signal is needed. Furthermore, to adapt to richer interactive scenarios, we intend to give other textile items the ability to perceive.

User-Centered Sensing. Our classification results suggested that per-user training has much higher accuracy than general pre-trained classifiers. This can be explained that the impedance properties of different users can be fairly unique. Furthermore, the



Figure 5: A series of envisioned applications for *eGlove*, for example, (a) the user can be reminded to take medicine; (b) *eGlove* could enable the children to learn new things; (c) the user can be gently reminded not to touch the nose and mouth to reduce the risk of infection

objects that people use are largely influenced by lifestyle and environment. To maximize the recognition robustness across more possible objects of our system, we intend to combine supervised and unsupervised learning methods to classify both general and automatically discover novel objects specific to individual users [22]. This manner reduces the burden to develop a user-specific model and also enhances the portability of wearable interactive fabrics for contextual interactions.

7 CONCLUSION

This work presents a wearable and low-cost Swept Frequency Capacitive Sensing system for objects and body part recognition on soft conductive fabrics. We illustrated the sensing principle and discussed our investigation on conductive fabric sensors. We built a prototype called *eGlove*, which was designed to maximize the sensitivity of the signal while preserving the essential properties of the fabric. By conducting a preliminary experiment, we found our approach demonstrated a 96.3% real-time classification accuracy with 10 daily necessities, 9 fruits and plants, and 6 body parts. We believe that our approach is meaningful and hope our work could benefit for future context-aware interactions with smart fabrics.

ACKNOWLEDGMENTS

This paper is supported by National Natural Science Foundation of China (52075478), Major Art project of National Social Science Foundation of China (20ZD09) and Major Project of Zhejiang Social Science Foundation (21XXJC01ZD).

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